Weekly Team Check-In 2: Evaluation & Testing

KPMG × Columbia QMSS Practicum (5052) · Since the midpoint presentation, Oct 26 – Nov 1, 2026

Workstream	Evaluation & Testing: measure whether the SEC-filings assistant's answers can be trusted
Team leads	Chuan-Yeu (Thomas) Song, Binyu Wang
Members and roles	Shenyi Feng (numeric accuracy) · Fei Huang (citation support) · Shuhan Bian (retrieval metrics; RAG liaison) · Yupin Wang (refusal/abstention) · Ruofei Duan (GraphRAG metrics; GraphRAG liaison) · Bo Yuan (latency, cost, consistency) · Binyu Wang (reasoning steps; Agents liaison) · Thomas Song (gold set; Data and KPMG liaison)
Where the work is	GitHub thomassong0604-oss/kpmg-eval → Week 6/ (code, data, results; 63 automated tests passing)

1. Progress since the previous check-in

Narrowed the gold set to the Data team's 10 companies

- Data screened its database (448 filings) and proposed **AAPL, CASY, MCK, MDT, MSFT, MU, NKE, NTAP, PH, TTWO** (21 filings: 10 10-Ks, 11 10-Qs). We rebuilt the gold set around them within a day.
- **Round-1 gold set: 139 questions** = 90 kept from Gold Set v1 + **49 new** questions on the added companies (34 lookups, 5 ten-company comparisons, 3 fiscal-calendar, 2 calculations, 5 unanswerable/false-premise). 10 questions that needed Amazon, Meta or NVIDIA were retired.
- Every new answer is computed in code from SEC XBRL data, **read twice: 53/53 values agree**, and **18/18** quarterly figures add up to the reported nine-month totals.
- Grader re-validated on the new questions: **156/156** right and wrong test answers labelled correctly (including negative values and scale errors).

Integrated the GraphRAG team's work

- Implemented GraphRAG's scoring requests from their eval-ask / eval-reply docs: citations now join on chunk, section or fact id; reruns compare on all three version fields.
- Wired both GraphRAG retrieval arms into our harness on their mock: **87 questions, 174 calls, 0 errors, 87/87 pass our contract checks**.
- Drafted **6 relationship questions** with GraphRAG (subsidiaries, segment changes, acquisitions), including **2 that test invented relationships**, the main way graphs fail. Shared a 155-question hop-count sheet and a one-page summary (Week 6/EVAL_GRAPHRAG.md).
- Found and fixed a bug in our own scoring: segment figures could count as company totals (27 false matches removed).

Worked with the RAG team

- Reviewed RAG's evidence anchors for our 100 questions: **97/99** numbers our answers use are covered; returned 2 non-matching and 33 over-broad anchors with fixes.
- RAG ran its retriever on our gold set (their report, 10/28): **recall@8 0.852** overall, 1.00 on segment questions, **0.69 on narrative** (weakest, consistent with our wording-robustness finding).

Decisions and deliverables

- Gold Set v1 stays frozen (file hash unchanged) so earlier scores remain comparable; Microsoft stays the MVP company; every cross-company question names each company's own fiscal period.
- Delivered: 5 Eval slides for the 10/29 KPMG meeting, Week 6 code and data on GitHub, Slack updates to Data, RAG and GraphRAG.

Feedback received and how we responded

Feedback	Response
GraphRAG: no question requires following a relationship; no check for invented links	6 relationship questions drafted, 2 test fabricated links; hops sheet shared for their input
GraphRAG: score graph answers by section/fact id, not chunk id	Done in code and tested
RAG: single-number anchors make recall too generous	Our review confirms it (33 broad anchors); fixes returned
RAG: add a fixed reader to measure numeric accuracy, report it by route	Accepted; next step 1 below

2. Current priorities

- **Numeric accuracy on RAG's real retriever** with the fixed reader RAG proposed (GPT-4o-mini, temperature 0), reported overall and by route (structured data vs. text). Shenyi Feng, Shuhan Bian, Bo Yuan.
- **Finishing the round-1 gold set:** evidence anchors for the 49 new questions once Data shares the filing text; human review of 8 flagged items. Thomas Song, Fei Huang, Yupin Wang.
- **GraphRAG and Agents integration:** agree a hop definition, review the relationship questions, agree the Agent's structured answer format (AnswerBlock with trace_id). Ruofei Duan, Binyu Wang.

3. Next steps before the next check-in

#	Action	Owner(s)	Done when
1	Plug RAG's fixed reader into the harness; score numeric accuracy on the 139-question set, overall and by route; log cost and latency per question	Shenyi Feng (lead), Shuhan Bian, Bo Yuan	Scorecard on GitHub and shared with RAG in Slack
2	Complete the round-1 gold set: anchors for the 49 new questions, review of unanswerable/false-premise items, the 10 retired questions removed from RAG's anchor file	Thomas Song (lead), Fei Huang, Yupin Wang	All 139 questions have verified anchors; status = reviewed
3	GraphRAG and Agent hand-offs: hop definition agreed and hops filled; harness pointed at GraphRAG's real retriever when ready; AnswerBlock agreed with Agents	Ruofei Duan (GraphRAG), Binyu Wang (Agents)	Hop values in hops_v1.csv; written AnswerBlock spec both teams accept

4. Challenges and blockers

- **Filing list not yet confirmed.** Data shared counts, not accession numbers. Our assumed list (each company's latest 10-K and 10-Q) reproduces their count exactly, but anchors could target the wrong filing if it differs. **Mitigation:** confirmation requested; rebuild is one script.
- **Six fiscal calendars.** The ten fiscal years end in six different months and the latest 10-Qs fall in three calendar quarters, so all ten cannot be compared on one quarter. **Mitigation:** every comparison question states each company's period.
- **Round 1 (W8) depends on parts still in progress:** GraphRAG's real backend (we tested on their mock) and the Agent's structured answers. If either is late, round 1 reports retrieval only for that component.
- **SEC data quirks** found while building questions: Micron's FY2026 10-K is not in scope, Parker-Hannifin reports net income under a different XBRL tag, Nike and NetApp changed their revenue tag. Each is now a test question, but they also affect the other teams' extraction.
- **Recall can look better than it is** when anchors are bare numbers that appear in several tables (RAG raised this too). Until anchors are fixed, we report recall with that caveat.
- **LLM cost:** the fixed reader and the two LLM-judge metrics need an API key and budget.
- **Team ramp-up:** eight members with mixed coding experience; everyone now needs to clone the repo and join the Slack channels. Thomas and Binyu are running setup help this week.

5. Cross-team coordination and support needed

From	Request
Data	Confirm the 21 accession numbers; share text chunks for all 21 filings; Exhibit 21 (subsidiaries) for the ten 10-Ks. Thank you for the screening work, which unblocked every team.
RAG	Fixed-reader spec and access; fix the 2 non-matching and 33 broad anchors; trace_id and cited chunk ids in the logs.
GraphRAG	Hop definition and values; a date for the real backend; 3 small fixes to the output format (units, segment names, unique ids).
Agents	Return the AnswerBlock (value, unit, period, citations, answered/refused) with trace_id.
KPMG	Confirm the 10-company scope; what accuracy level counts as "good enough" for a numeric answer.
Instructor / TAs	Guidance on an API budget for the fixed reader and LLM-judge metrics; whether round 1 may report retrieval-only scores for components not ready by W8.